PATTERNS:

1.FULL PYRAMID:

```
public class pattern{
public static void main(String[]args){
int n=5;
for(int i=1;i<=n;i++){
for(int j=i;j<n;j++){
  System.out.print(" ");
  }
  for(int k=1;k<=(2*i-1);k++){
     System.out.print("*");
  }
  System.out.println();
 }
}
```

2.INVERTED FULL PYRAMID:

```
public class inverted{
public static void main(String[]args){
 int n=5;
   for(int i=n;i>=1;i--){
     for(int j=n;j>1;j--){
     System.out.print("");
     for(int k=1;k<=(2*i-1);k++)
     {
     System.out.print("*");
     }
   }
   System.out.println();
 }
}
}
```

3.DIAMOND PYRAMID:

```
public class diamond{
public static void main(String[]args){
  int n=5;
  for(int i=1;i<=n;i++)
  {
```

```
        for(int j=i;j<n;j++){
          System.out.print(" ");
        }
        for(int k=1;k<=(2*i-1);k++){
          System.out.print("*");
        }
        System.out.println();
      }
      for(int i=n-1;i>=1;i--){
        for(int j=i;j<n;j++){
          System.out.print(" ");
          }
        for(int k=1;k<=(2*i-1);k++){
          System.out.print("*");
        }
        System.out.println();
      }
    }
}
```

4.HOLLOW PYRAMID:

```
  public class hollow{
  public static void main(String[]args){
    int n=5;
    for(int i=1;i<=n;i++){
      for(int j=1;j<=n;j++){
        System.out.print(" ");
      }
      for(int j=1;j<=2*i-1;j++){
        if(j==1 ||j==2*i-1||i==n){
          System.out.print("*");
                }else{
                  System.out.print(" ");
                }
      }
      System.out.println();
    }
  }
}
```

5.INVERTED HOLLOW PYRAMID:

```
public class invhollow{
 public static void main(String[]args){
   int n=5;
   for(int i=1;i>=n;i--){
     for(int j=1;j<=n-i;j++){
       System.out.print(" ");
     }
     for(int j=1;j<=2*i-1;j++){
       if(i==n||j==1||j==2*i-1){
         System.out.print("*");
                 }else{
                   System.out.print(" ");
                 }
     }
     System.out.println();
   }
 }
```

6.HOURGLASS PYRAMID:

```
public class hourglass{
public static void main(String[]args){
   int n=5;
   for (int i = n; i >= 1; i--) {
     for (int j = 0; j < n - i; j++) {
       System.out.print(" ");
     }
     for (int j = 1; j <= 2 * i - 1; j++) {
       System.out.print("*");
     }
     System.out.println();
   }
   for (int i = 2; i <= n; i++) {
     for (int j = 0; j < n - i; j++) {
       System.out.print(" ");
     }
     for (int j = 1; j <= 2 * i - 1; j++) {
       System.out.print("*");
     }
     System.out.println();
   }
 }
}
```

7.BUTTERFLY PYRAMID:

```
  public class butterfly{
  public static void main(String[]args){
    int n=5;
    for (int i = 1; i <= n; i++) {
      for (int j = 1; j <= i; j++) {
        System.out.print("*");
      }

      for (int j = 1; j <= 2 * (n - i); j++) {
        System.out.print(" ");
      }

      for (int j = 1; j <= i; j++) {
        System.out.print("*");
      }

      System.out.println();
    }
     for (int i = n; i >= 1; i--) {
      for (int j = 1; j <= i; j++) {
        System.out.print("*");
      }

      for (int j = 1; j <= 2 * (n - i); j++) {
        System.out.print(" ");
      }

      for (int j = 1; j <= i; j++) {
        System.out.print("*");
      }

      System.out.println();
    }
  }
}
```